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Clinical Feature Construction & Selection with Agentic AI

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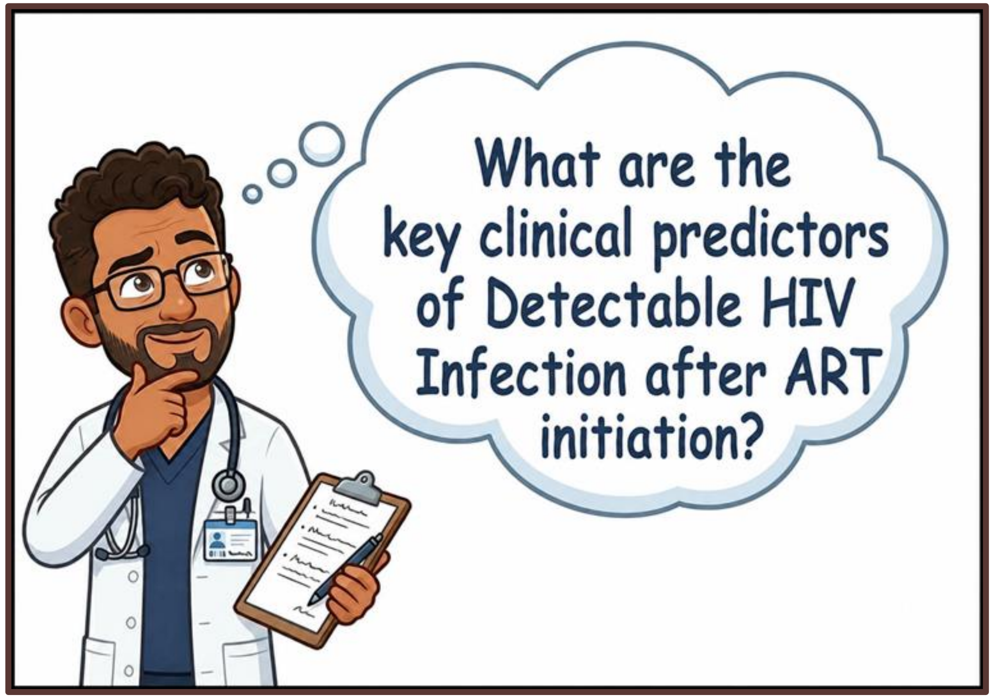
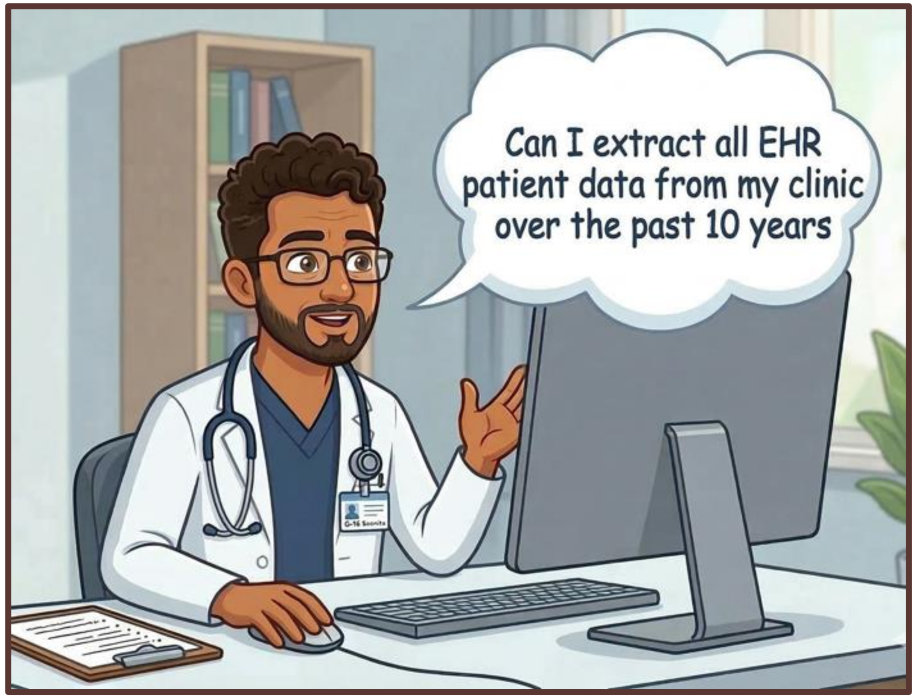
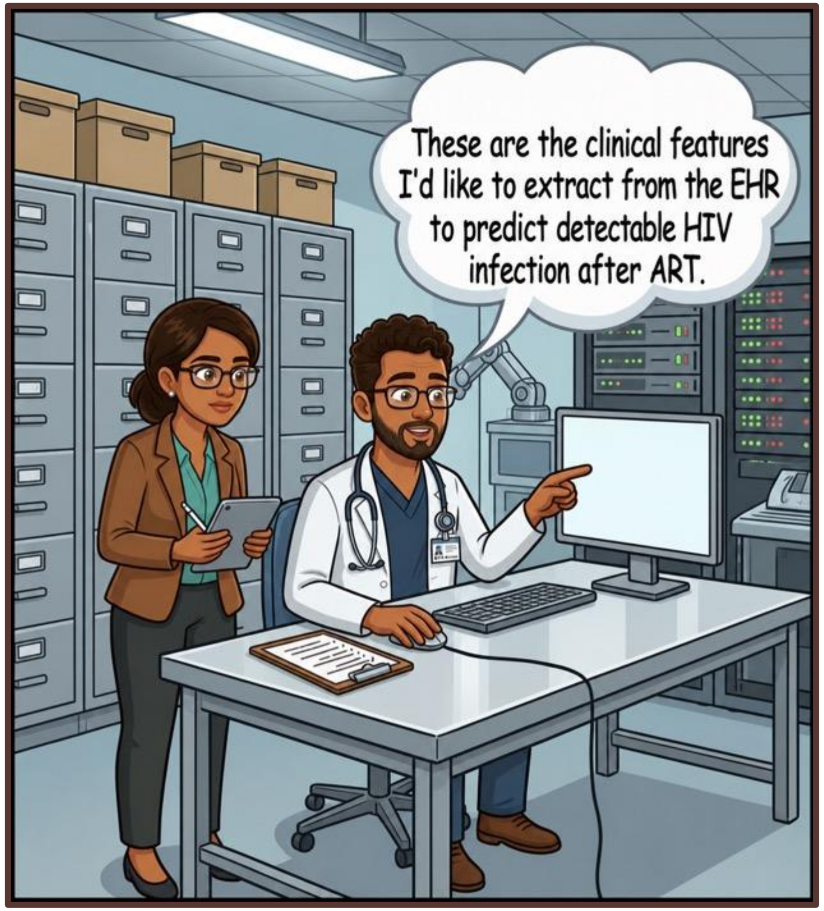
Step 1: Develop a Hypothesis and Pull Data





Step 2: Feature Construction



The Question

Patient ID	ICD-10 codes (comma-separated)
827969	M54.30,Z00.00,Z00.00,M54.32,M47.892,M48.061,M51.26,M51.16,Z01.818,
253361	K21.9,M54.50,M54.50,M48.061,M43.10,M48.061,M43.06,Z01.818,M54.9,I4
531301	N20.0,N20.0,K40.90,Z98.890,R59.9,N28.89,R10.9,R19.7,I10,R11.2,R14.3,
667755	K21.9,R07.89,R06.02,R51,K21.9,R07.89,N40.0,H52.209,Z86.69,H52.00,H5
59480	K64.4,M51.37,M21.379,M51.26,M21.372,M51.26,M51.16,F41.9, F32.A,J30.
618058	H93.13,E66.3,Z00.00,M79.606,Z23,M25.569,Z12.11,M43.10,Z12.11,E78.00
61444	I26.99,I26.99,I26.99,I26.99,T24.221A,T24.221A,T24.221A,T24.221A,L03.11
389040	F16.11,R11.2, R19.7,F10.10,F12.11,F13.90,F40.01,R10.13,F10.20,F33.1,F4
421287	I50.42,I50.42,I50.42,I50.42,J42,J42,J42,J42,J42,J42,J42,J42,I25.10,I25.10,I2
155786	K57.30,J44.9,G89.29,F17.200,M47.819,R20.2,K64.9,E78.5,G63,R73.03,E78
109184	J40,R09.82,J40,R09.82,M43.16,N95.2,N95.2,N89.8,K63.5,M54.50, G89.29,
344143	N91.1,N91.1,Z23,M41.9,M41.9,N91.2,N91.2,M41.25,M41.20,M41.25,M62.
857905	Z72.0,Z72.0,G83.14,G83.14,R53.1,R53.1,M54.16,M54.16,M48.061,M48.06
623898	M54.50,M71.38,R93.7,M54.41, M54.42, G89.29,M25.552,M79.604, M79.60
522536	Z87.891,M25.519,M19.039,M06.9,Z79.899,Z11.1,M06.9,M19.079,L84,M20.
200174	E78.5,R73.09,M54.50,M54.50,Q05.9,M54.50,M16.12,E78.00,M16.12,M51.
38483	K21.9,K21.9,K21.9,K21.9,M51.37,M51.37,M51.37,M51.37,M47.817,M47.81
422350	IMO0002,Z01.818,K59.09,I10,E78.5,A41.9, R65.21,E86.0,R10.84,D64.9,I10
175015	M25.519,M25.519,N12,N12,I60.9,S53.104A,V29.9XXA,R41.82,I60.9,I49.9,V

Can Feature Construction be Fully Automated?

Goals

Produce a fully autonomous, agentic system to:

1. Identify, construct, and select relevant EHR features (ICD-10 codes, labs, medications, vitals, and demographics) predictive for a HIV viral load status (detectable vs undetectable)

2. Create a dataset for downstream predictive modeling

What we have at our disposal:

Knowledge – Curated Sources of Biomedical Knowledge

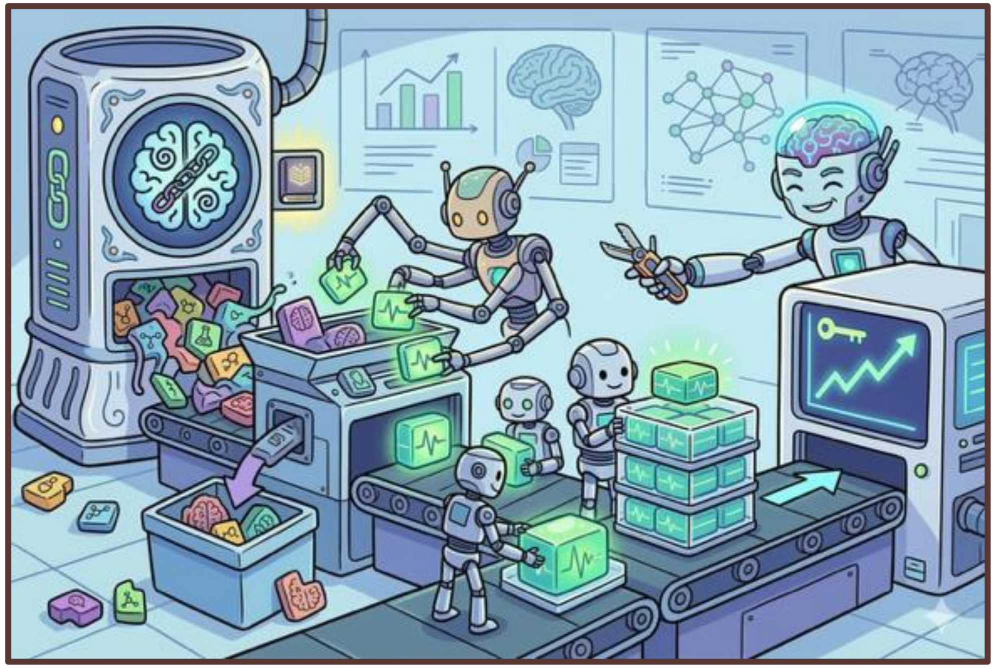
- AddictionKB: addiction, psychiatric comorbidity, and HIV knowledge base
- Summaries of 300+ relevant papers in Open Knowledge Format (OKF)

Inference Model

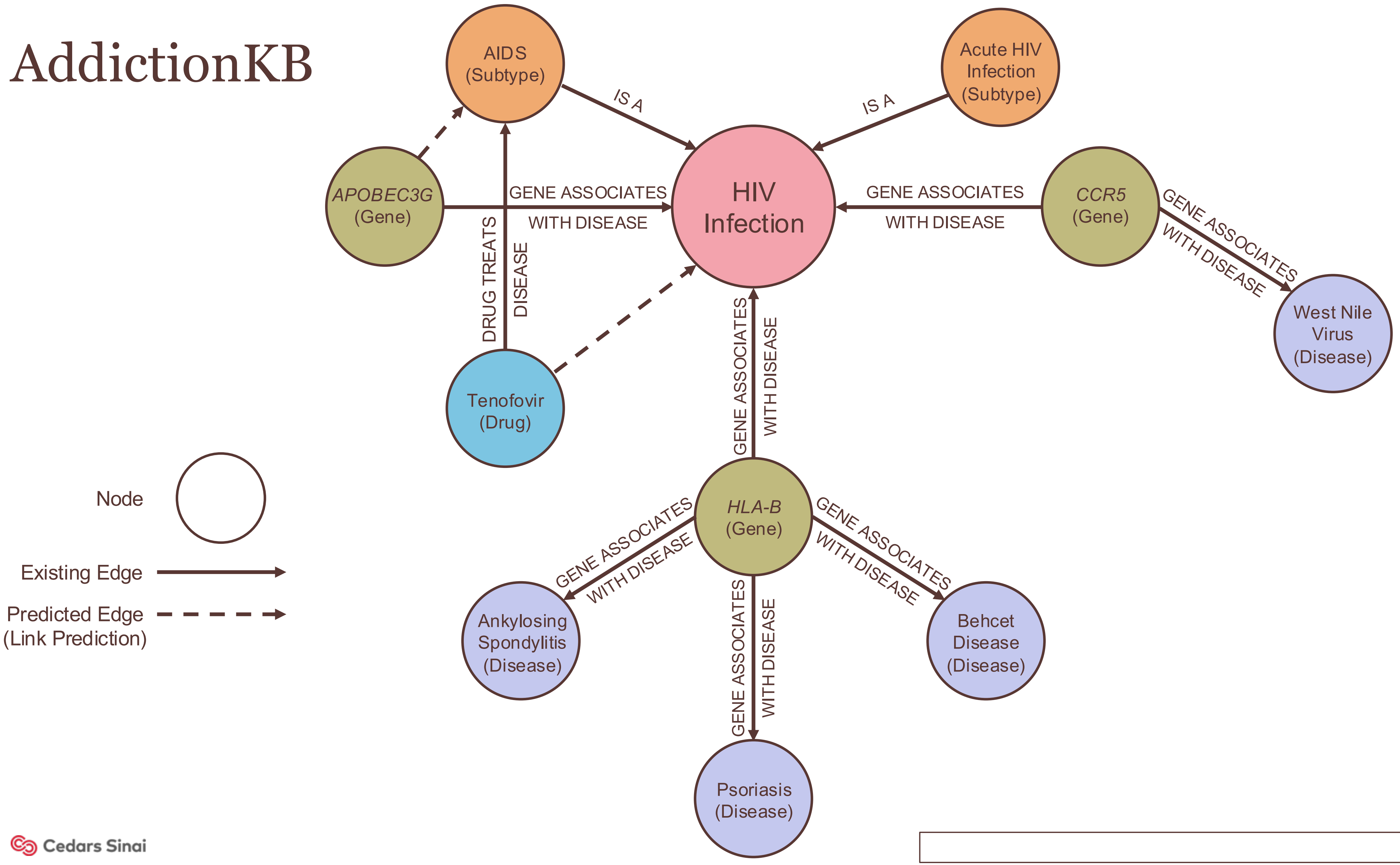
- Claude Sonnet 5

Tools

- Relevant software packages and scripts, including a semantic matcher
  - Semantic matcher aligns knowledge graph entities, OKF terms, and model inference entities (thoughts) to EHR data entities for proper feature construction



AddictionKB



Optimizing Your Party: The Value of Diverse Agentic Roles

The Agentic Feature Construction System

Autonomous Agents & Their Roles

Steps 1 & 7: Architect – Supervisor
- Initiates the linear agentic graph
- Makes final decisions on feature inclusion/exclusion for feature construction

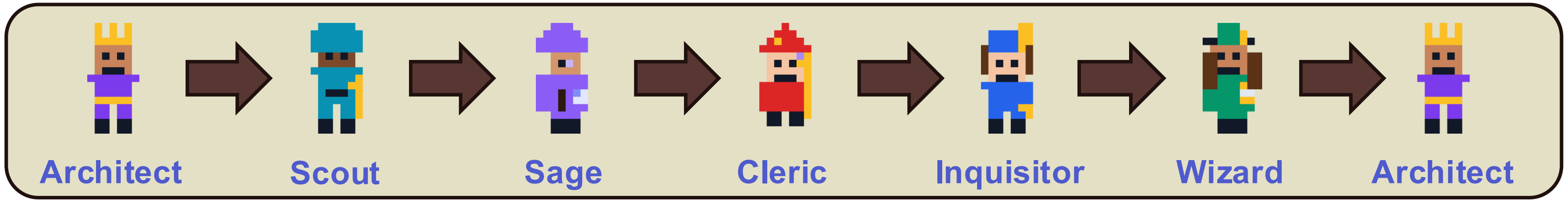
Step 2: Scout – Knowledge Graph (KG) Explorer ⚒
- Traverses the KG for clinical relationships
  - 1 hop to *n*-hop traversal
- Performs link prediction to discover novel associations

Step 3: Sage – Literature Expert (OKF) ⚒
- Navigates relevant clinical knowledge (papers and abstracts) to identify potential features
- Uses the Open Knowledge Framework (OKF)

Step 4: Cleric – Clinician
- Decides what clinically relevant concepts are missing from the KG and OKF
- Uses expertise to suggests new features

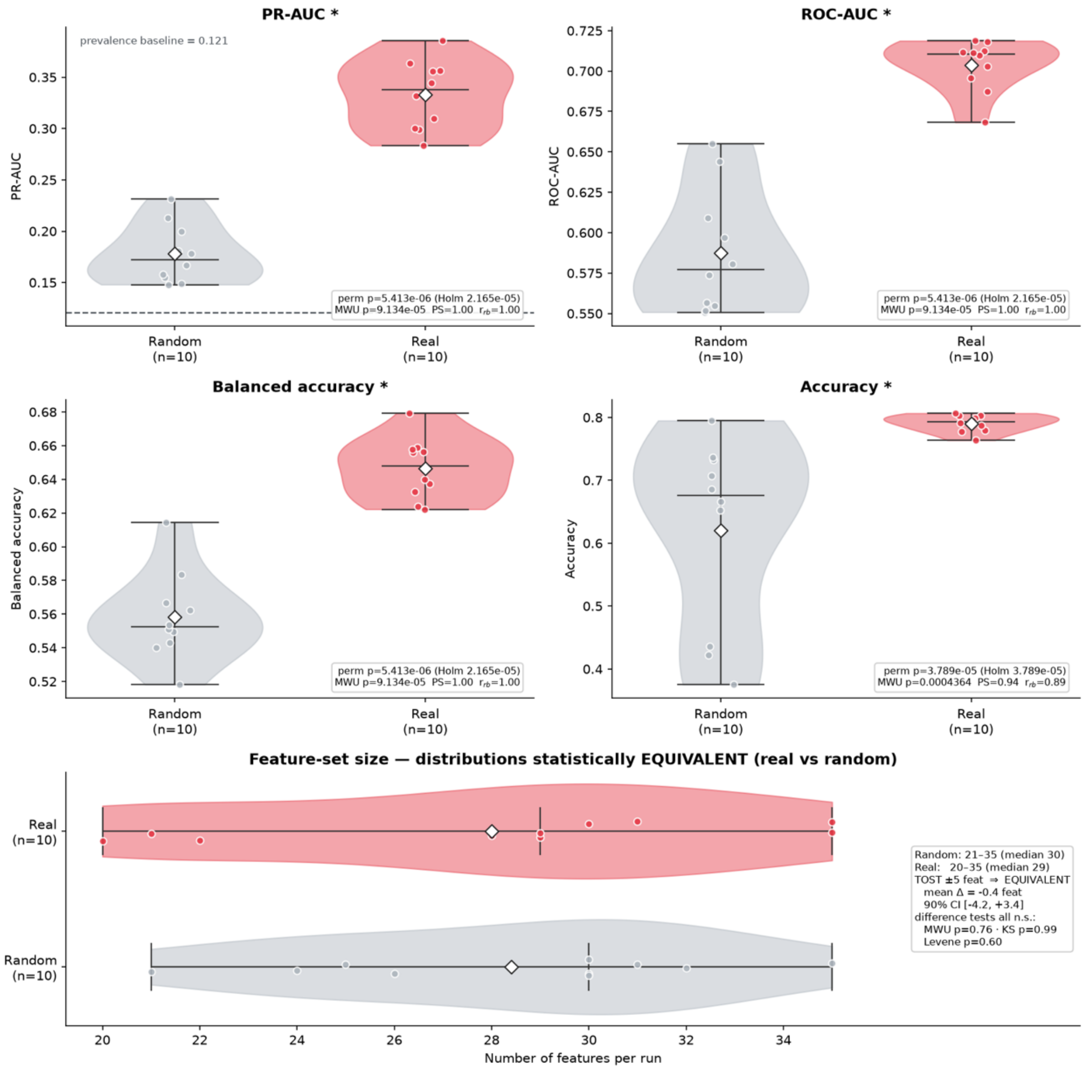
Step 5: Inquisitor – Critic
- Critiques all features provided by agents 2-4 with internal knowledge
- Provides feedback and suggestions for inclusion/exclusion

Step 6: Wizard – Statistician & Feature Selector ⚒
- Constructs all pending features in a temporary dataframe and performs unsupervised statistical analyses
  - Assesses missingness, variances, correlations, & prevalences
- Provides feedback and suggestions for inclusion/exclusion
- Loops back to Architect for final inclusion/exclusion decisions

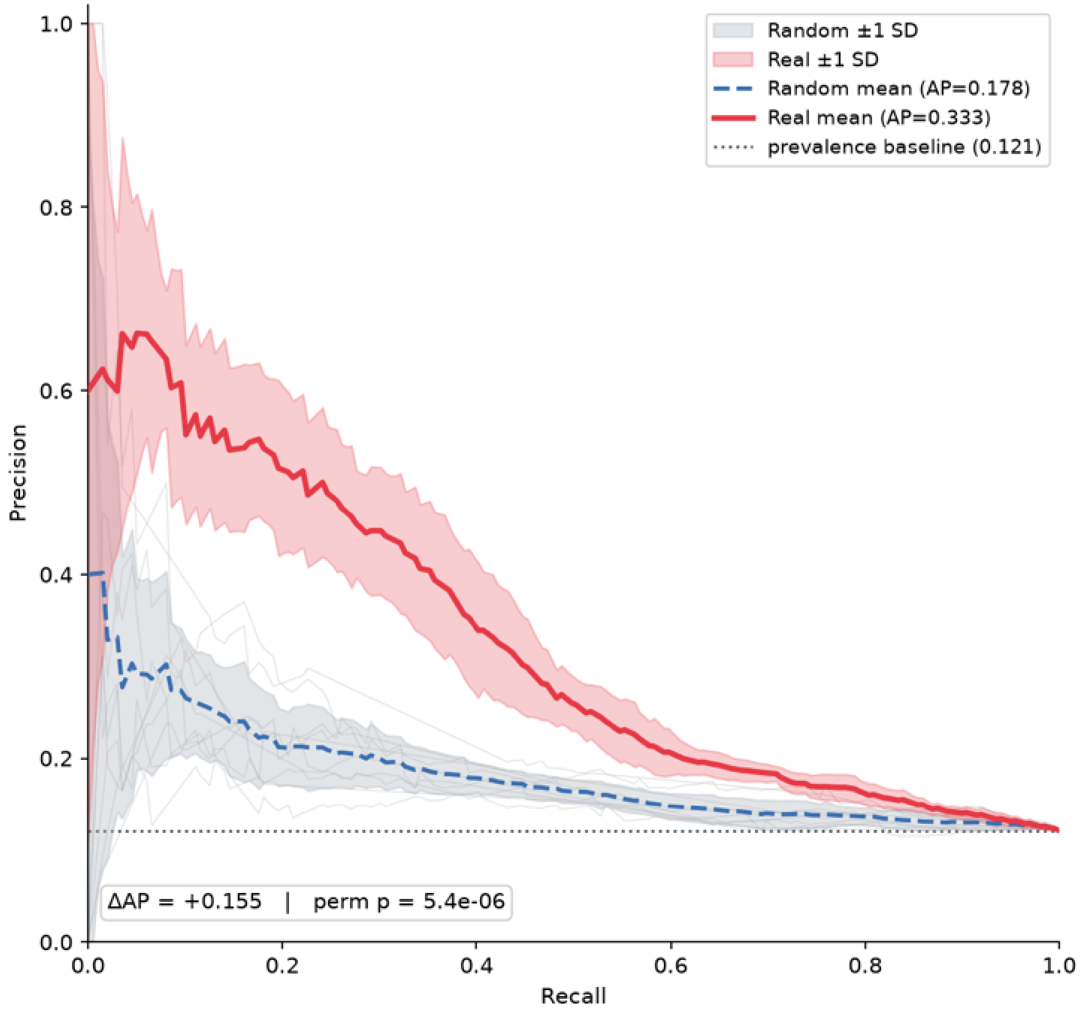


Results – Predictive Metric Comparisons to Random

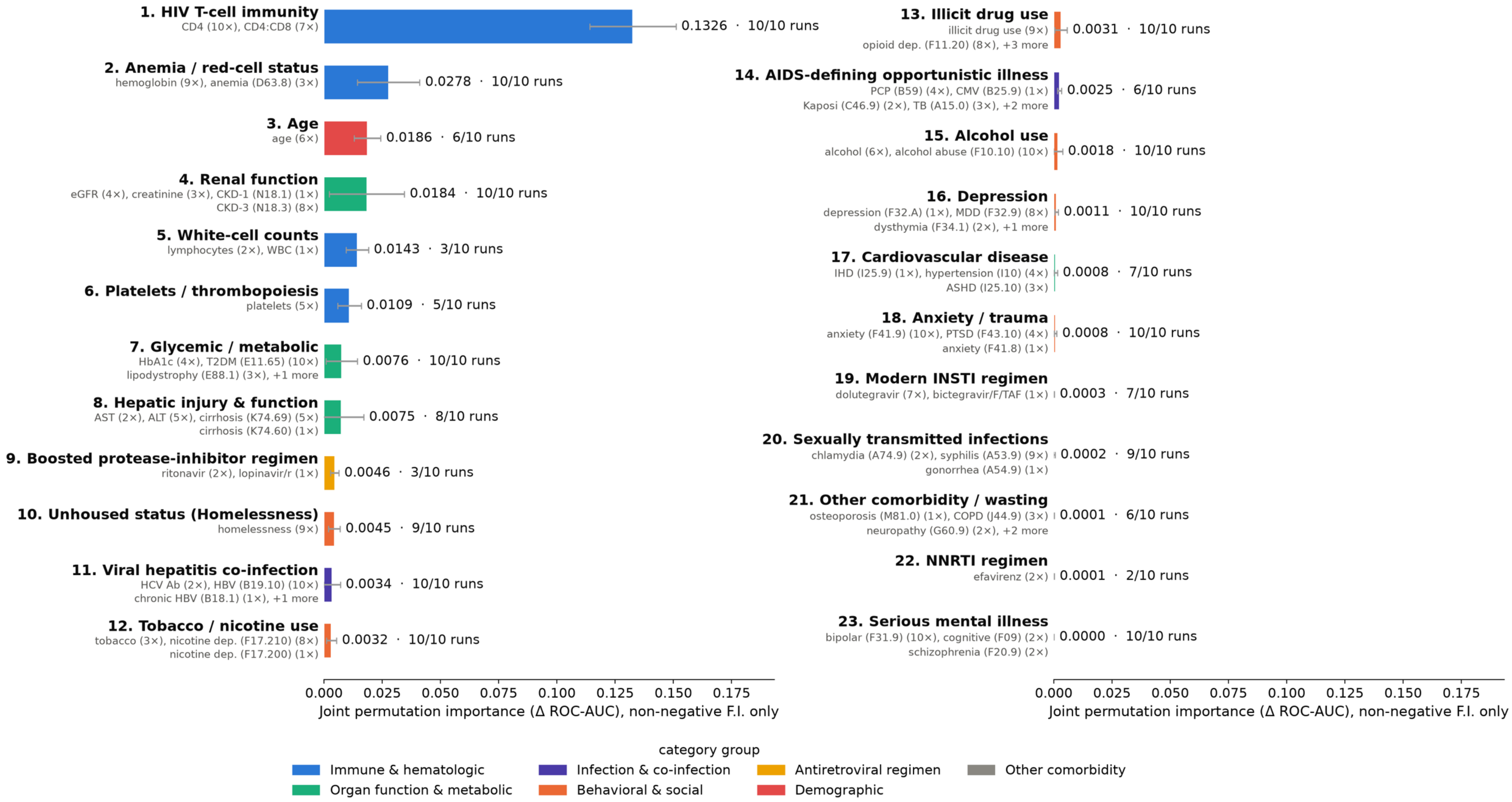
Agentic features vs. random baseline features (same seed [42] split)



Precision-Recall: Agentic Features vs Random Baseline Features

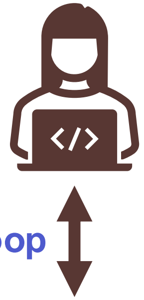


Results – Aggregate Feature Importance by Clinical Category



Future Agents & Architectural Improvements

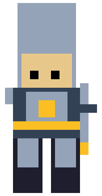
1. Human-in-the-Loop



2. Complex Graph and Looping Topologies

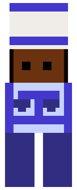
3. Topology/LLM Optimization with Evolutionary Algorithms

4. Expansion Pack Agents



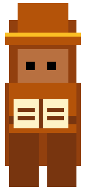
Paladin – Optimization and QA

- Optimizes pipeline hyperparameters/topology
- Assess tool use and code quality



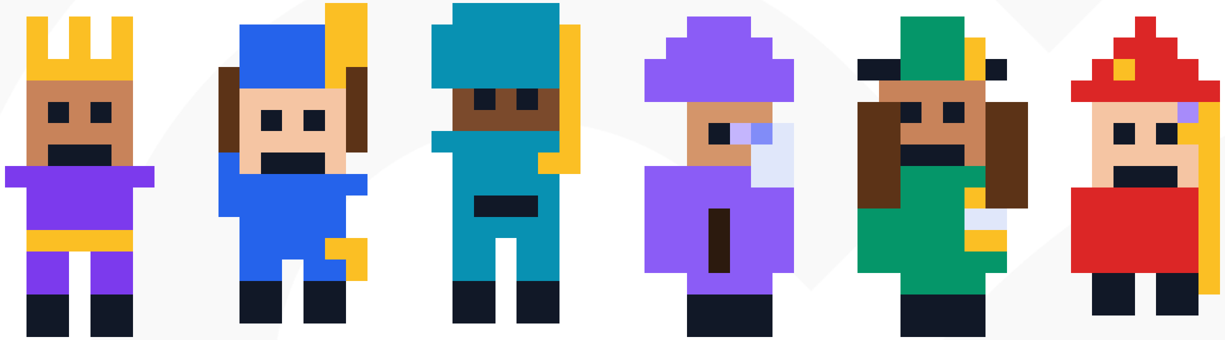
Justicar – AI Fairness/Replicability

- Assesses feature/cohort bias/fairness
- Suggests alternative sampling/features
- Tests pipeline replicability



Farseer – Feature Engineering

- Creates new meta features using logical operators or by clinical insight via OKF/KG



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